

Adaptive Kernel Selection for Stein Variational Gradient Descent

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Abstract

A central computational problem in Bayesian inference is the efficient approximation of intractable posterior distributions. Stein Variational Gradient Descent (SVGD) is a deterministic variational inference procedure that transports a finite ensemble of interacting particles so that their empirical measure approximates a target distribution. Its dynamics are governed by the geometry of a reproducing kernel Hilbert space (RKHS) and are, consequently, strongly dependent on the choice of the kernel function, which controls both the rate of convergence and the fidelity of the resulting approximation. The widely adopted median heuristic prescribes a single scalar bandwidth in closed form; it is computationally inexpensive but structurally inflexible, and its approximation quality degrades in high-dimensional regimes. In this work, we formulate an alternative procedure for selecting kernel parameters adaptively over an abstract parameterized family of kernels. Recent convergence analyses expressed in terms of the kernelized Stein discrepancy (KSD) indicate that selecting kernel parameters so as to maximize the KSD yields improved performance. Building on this observation, we introduce Adaptive SVGD (Ad-SVGD), an alternating scheme that interleaves particle transport via SVGD with adaptive kernel-bandwidth updates obtained by gradient ascent on the KSD. We develop a simplified convergence analysis that extends existing guarantees for KSD minimization under a fixed kernel to the adaptive setting, establishing that the maximal KSD over the parameterized kernel class vanishes along the iterates. These theoretical findings are corroborated empirically: Ad-SVGD consistently attains lower approximation error than standard heuristics across a range of inference tasks.

1 Introduction

Stein Variational Gradient Descent (SVGD) [18] is a deterministic, particle-based algorithm for approximate Bayesian inference that has been widely adopted as an alternative to sampling-based Markov Chain Monte Carlo (MCMC) methods. At each iteration, SVGD advances a finite ensemble of particles along an update direction obtained from the functional gradient of the Kullback-Leibler (KL) divergence to the target distribution, subject to the constraint that the direction lie in the unit ball of a reproducing kernel Hilbert space (RKHS). A principal limitation of the procedure is its acute sensitivity to the kernel, whose choice governs the inter-particle interaction and thereby the algorithm’s empirical performance [8, 28]. A further well-documented deficiency is that the resulting particle approximation systematically underestimates the posterior variance [2]. Taken together, these observations have motivated the prevailing view that SVGD degrades as the dimension of the state space grows. We contest this view by introducing an adaptive mechanism for kernel-parameter selection that tunes the kernel online during inference by maximizing the kernelized Stein discrepancy (KSD), thereby inducing more effective transport in complex, high-dimensional target distributions.

1.1 Related work.

Since its introduction [18], SVGD has been deployed as a general-purpose tool for approximate Bayesian inference across numerous machine learning applications [22, 24, 30, 15]. A substantial body of subsequent work has characterized its theoretical behavior. Mean-field convergence has been established in both the continuous-time [23, 8, 5] and discrete-time [16, 32] regimes, and finite-particle convergence rates have been derived under various regularity assumptions [3, 33]. The dynamics have furthermore been identified with a gradient flow on the space of probability measures [19, 8], in analogy with the variational formulation of Jordan, Kinderlehrer, and Otto [14].

The performance of SVGD is critically determined by the kernel, which specifies both the pairwise particle interaction and the convergence behavior of the iteration (see also Figures 1 and 9). Convergence guarantees are typically stated as mean-field convergence in the KSD metric, whose relationship to weak convergence is itself kernel-dependent [11]. The prevalent median heuristic [12] admits a trivial implementation but is devoid of theoretical justification and exhibits provable performance degradation as the problem dimension increases [2, 40, 36]. A line of subsequent work has proposed mechanisms to counteract this high-dimensional degradation [7, 37, 10, 21], and a complementary line replaces the closed-form update direction with one parameterized and learned by neural networks [17, 39].

The most closely related approach is that of Ai et al. [1], who develop a mixture-of-kernels framework: they define a KSD associated with a weighted linear combination of a finite set of base kernels and learn the mixture weights by maximizing this multiple-kernel KSD. Their formulation is, however, restricted to a finite-dimensional kernel basis and does not address optimization over a continuous kernel-parameter space, which is the object of the present work.

1.2 Contributions.

We investigate the following research question: can the unmodified SVGD iteration, without architectural or algorithmic augmentation, attain competitive performance when equipped with a principled, adaptively selected kernel? By isolating kernel selection as the sole degree of freedom and holding all other components of the base algorithm fixed, we seek to quantify the extent to which kernel choice alone determines the effectiveness of SVGD and to delineate the limits of what adaptive kernel learning can achieve within the original dynamics.

Our main contributions are as follows.

- (i) **Adaptive Kernel Selection Method.** We introduce a procedure that updates the kernel parameters online by maximizing the KSD over the course of SVGD inference. In contrast to the median heuristic, which is confined to a single scalar bandwidth, our formulation optimizes a vector of continuous kernel parameters, affording substantially greater expressivity and adaptivity in the particle updates.
- (ii) **Theoretical Analysis.** We supply theoretical justification by analyzing the algorithm in the discrete-time mean-field setting, extending existing fixed-kernel convergence results for SVGD. Concretely, we prove that the supremum of the KSD over the parameterized kernel class converges to zero as the particle measure approaches the target. Under an additional Stein logarithmic Sobolev inequality, we further derive explicit iteration-complexity bounds in the mean-field limit.

- (iii) **Empirical Validation.** Through numerical experiments, we establish that the proposed adaptive kernel selection consistently outperforms the median heuristic and mitigates the variance-collapse phenomenon.

2 Mathematical background

Throughout, let $k : \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$ denote a symmetric positive-definite kernel and let $\mathcal{H}_0$ be its associated RKHS. We define $\mathcal{H}$ to be the d -fold Cartesian product of $\mathcal{H}_0$, endowed with the inner product $\langle f, g \rangle_{\mathcal{H}} = \sum_{i=1}^d \langle f_i, g_i \rangle_{\mathcal{H}_0}$, and we write $\Phi_k(x) = k(\cdot, x) \in \mathcal{H}_0$ for the canonical feature map. We denote by $x \cdot y$ the Euclidean inner product on $\mathbb{R}^d$ and by $\nabla \cdot$ the divergence operator. The set of probability measures on the measurable space $(\mathbb{R}^d, \mathcal{B}(\mathbb{R}^d))$ is written $\mathcal{P}(\mathbb{R}^d)$, and $\mathcal{P}_p(\mathbb{R}^d)$ denotes the subset of measures with finite p -th moment. For $\mu, \nu \in \mathcal{P}_p(\mathbb{R}^d)$, we define the Wasserstein p -distance

$$\mathcal{W}_p(\mu, \nu) = \inf_{\gamma \in \Gamma(\mu, \nu)} \left(\int_{\mathbb{R}^d \times \mathbb{R}^d} \|x - y\|_2^p d\gamma(x, y) \right)^{1/p},$$

where $\Gamma(\mu, \nu)$ is the set of couplings of μ and ν , i.e. the set of probability measures on $\mathbb{R}^d \times \mathbb{R}^d$ with marginals μ and ν .

2.1 Integral probability metrics and kernelized Stein discrepancy.

Integral probability metrics (IPMs) [26] quantify the discrepancy between two measures as the supremum, over a prescribed class of test functions, of the difference between their expectations. Rendering this construction tractable for measuring the distance from a sample to an intractable target requires a test-function class whose elements integrate to zero against the target; Stein's method [35] furnishes precisely such a class. Concretely, for the Stein operator $\mathcal{S}_\pi f := \nabla \log \pi \cdot f + \nabla \cdot f$ and an appropriate kernel, one has the zero-mean identity $\int_{\mathbb{R}^d} \mathcal{S}_\pi f(x) d\pi(x) = 0$ for every $f \in \mathcal{H}$ [6, 20]. The induced *kernelized Stein discrepancy* (KSD) is defined by

$$\text{KSD}(\mu|\pi) := \sup_{f \in B(\mathcal{H})} \left| \int_{\mathbb{R}^d} \mathcal{S}_\pi f d\mu \right|, \quad (1)$$

where $B(\mathcal{H})$ denotes the closed unit ball of $\mathcal{H}$. This variational problem admits the closed-form maximizer $f^* = \frac{\psi}{\|\psi\|_{\mathcal{H}}}$, with $\psi = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$ and $\mathcal{A}_\pi^k(x) = \nabla \log \pi(x) \Phi_k(x) + \nabla \Phi_k(x) \in \mathcal{H}$, where Φ_k is the feature map of the kernel k . Substituting the maximizer, the supremum reduces to the RKHS norm of ψ , yielding the equivalent characterization $\text{KSD}(\mu|\pi) = \|\psi\|_{\mathcal{H}}$.

2.2 Stein Variational Gradient Descent.

Given a target π and a reference measure μ_0 , SVGD constructs an approximation of π by the recursion $\mu_{n+1} := T_{\sharp} \mu_n$, where $T_{\sharp}$ is the push-forward operator associated with the transport map $T(x) = x + \gamma \psi^{\mu_n}(x)$ and the vector field $\psi^{\mu} = \int_{\mathbb{R}^d} \mathcal{A}_\pi^k d\mu$, defined for $\mu \in \mathcal{P}(\mathbb{R}^d)$, is the direction of steepest descent. This choice is dictated by the maximizer of Equation (1): the normalized field $\frac{\psi^{\mu_n}}{\|\psi^{\mu_n}\|_{\mathcal{H}}}$ minimizes the directional derivative $\left. \frac{d}{d\gamma} \text{KL}(T_{\sharp} \mu_n \| \pi) \right|_{\gamma=0}$ over the unit ball of $\mathcal{H}$ [cf. 18, Theorem 3.1], where $\text{KL}(\cdot \| \cdot)$ denotes the KL divergence. For this choice, the directional derivative

equals the negative squared KSD:

$$\frac{d}{d\gamma} \text{KL} \left(\left(\text{Id} + \gamma \psi^{\mu_n} \right)_{\#} \mu_n \parallel \pi \right) \Big|_{\gamma=0} = -\text{KSD}^2(\mu_n | \pi). \quad (2)$$

Applying the maps T generated in this manner iteratively to a particle ensemble $\{X_0^i\}_{i=1}^M$ drawn from μ_0 , and replacing μ_n by the empirical measure of the particles, yields the coordinate update rule

$$X_{n+1}^i = X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k(X_n^i, X_n^j). \quad (3)$$

3 Adaptive kernel selection for SVGD

Because establishing convergence of SVGD in the KL divergence is analytically difficult and demands restrictive hypotheses, recent analyses have instead quantified convergence through the kernelized Stein discrepancy (KSD) [16, 33, 32]. Minimization of the KSD is, however, insufficient to guarantee weak convergence: the associated sequence of mean-field measures need not be tight, and tightness is a necessary condition for weak convergence of probability measures on a Polish space. Furthermore, as the variational problem Equation (1) makes explicit, the RKHS structure—hence the kernel—controls the degree to which convergence in KSD implies weak convergence [11].

To address this deficiency and thereby strengthen the convergence guarantees, we formulate an adaptive variant of SVGD over a parameterized kernel family $\{k_\theta \mid \theta \in \Theta\}$. For each parameter $\theta \in \Theta$ we write KSD_θ for the corresponding *kernelized Stein discrepancy*, ψ_θ^μ for the associated optimal update direction, and Φ_θ for the associated feature map. This explicit parameterization renders the kernel a decision variable that is selected adaptively during optimization, allowing the procedure to exploit the kernel’s influence on the convergence properties and to overcome the rigidity inherent in any fixed choice.

The design principle underlying our approach is to select, at each step, the kernel that maximizes the KSD between the empirical particle measure and the target. The rationale is direct: by Equations (2) and (7), the instantaneous decrease of the KL divergence along the SVGD flow equals the squared KSD evaluated at the current particle measure. Consequently, maximizing the KSD over the kernel class at a given iterate is equivalent to maximizing the instantaneous rate of KL decrease. This principle is corroborated by the gradient-flow interpretation of SVGD, under which a kernel inducing a larger KSD yields a strictly more favorable convergence rate when the associated RKHSs are compared [28, 8].

Whereas standard SVGD seeks convergence of the KSD for a single fixed kernel, the proposed algorithm targets the supremum of the KSD over the entire kernel class. Unlike the median heuristic of Liu and Wang [18], the procedure applies to an arbitrary parameterized kernel family: at each iteration it computes an optimal parameter

$$\theta_n \in \underset{\theta \in \Theta}{\operatorname{argmax}} \text{KSD}_\theta(\mu_n | \pi).$$

We realize this objective by advancing the kernel parameter with one or more gradient-ascent steps on the KSD, using a step size $s > 0$, prior to executing the particle update; to control runtime, we additionally permit the parameter to be held fixed on a subset of iterations rather than updated at every step (see Algorithm 1). This behavior is governed by a user-specified Boolean

decision variable paramupdate. We remark that the base transport step admits substitution by more sophisticated variants (e.g., adaptive step-size schedules, line searches, momentum, or second-order/preconditioned updates) without any modification to the kernel-selection mechanism, and that the gradient-ascent update on θ may likewise be replaced by an alternative optimization scheme. Our implementation of the kernel update is based on the closed-form expansion [6, 20]

$$\text{KSD}^2(\mu|\pi) = \int_{\mathbb{R}^d \times \mathbb{R}^d} u_\pi^k d(\mu \otimes \mu) \quad (4)$$

with

$$u_\pi^k(x, y) = k(x, y) \nabla \log \pi(x) \cdot \nabla \log \pi(y) + \nabla \log \pi(y) \cdot \nabla_x k(x, y) \\ + \nabla \log \pi(x) \cdot \nabla_y k(x, y) + \text{trace}(\nabla_x \nabla_y k(x, y)), \quad (5)$$

from which the gradient with respect to θ is obtained by direct differentiation. Two observations characterize the incremental computational cost of Ad-SVGD relative to standard SVGD:

- The gradient-ascent steps on the kernel parameter reuse the score evaluations $\nabla \log \pi(X_n^i)$ already required by the corresponding SVGD step; consequently, Ad-SVGD incurs no additional evaluations of $\nabla \log \pi$.
- The residual overhead can be further curtailed by limiting the number of parameter-update steps `nstepstheta`, or by updating the kernel parameter only once every several SVGD steps. In the large-ensemble regime, i.e. when M is large, the KSD may be estimated from a subsampled subset of particles.

Algorithm 1: Ad-SVGD

Input: Initial particle set $\{X_0^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$, kernel class $\{k_\theta \mid \theta \in \Theta\}$, initial kernel parameter $\theta_{-1} \in \Theta$, step sizes $\gamma, s > 0$, number of steps `nsteps`, `nstepstheta` $\in \mathbb{N}$

Output: Final particle set $\{X_{\text{nsteps}}^i \in \mathbb{R}^d \mid i = 1, \dots, M\}$

```

for  $n = 0$  to nsteps  $- 1$  do
  Decide paramupdate  $\in \{\text{True}, \text{False}\}$ ;
  if paramupdate then
     $\theta_n^0 \leftarrow \theta_{n-1}$ ;
    for  $\ell = 0$  to nstepstheta  $- 1$  do
       $\theta_n^{\ell+1} \leftarrow \theta_n^\ell + s \nabla_{\theta_n^\ell} \text{KSD}_{\theta_n^\ell}^2 \left( \frac{1}{M} \sum_{i=1}^M \delta_{X_n^i} \mid \pi \right)$ ;
    end
     $\theta_n \leftarrow \theta_n^{\text{nstepstheta}}$ ;
  else
     $\theta_n \leftarrow \theta_{n-1}$ ;
  end
  for  $i = 1$  to  $M$  do
     $X_{n+1}^i \leftarrow X_n^i + \frac{\gamma}{M} \sum_{j=1}^M k_{\theta_n}(X_n^i, X_n^j) \nabla \log \pi(X_n^j) + \nabla_{X_n^j} k_{\theta_n}(X_n^i, X_n^j)$ ;
  end
end

```

4 Convergence analysis

We consider a target measure admitting a Lebesgue density of Gibbs form $\pi(x) \propto \exp(-V)$, where $V : \mathbb{R}^d \rightarrow \mathbb{R}$ is a potential. To furnish theoretical support for Ad-SVGD, we extend recent KSD-based convergence results for SVGD to the adaptive setting. Specifically, we generalize the analysis of [32], which is carried out under Talagrand’s inequality—a condition that holds under mild hypotheses on the target and is strictly weaker than the frequently invoked logarithmic Sobolev inequality. We impose the assumptions of [32], required now to hold uniformly over all kernel parameters $\theta \in \Theta$.

Assumption 1. *The potential satisfies $V \in C^2$ with $\int_{\mathbb{R}^d} \exp(-V(x)) dx < \infty$, and its Hessian H is uniformly bounded in the operator norm; that is, there exists a constant $L \geq 0$ such that $\|H(x)\|_{\text{op}} \leq L$ for every $x \in \mathbb{R}^d$.*

Assumption 2. *The target $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies Talagrand’s transport inequality T1: there exists a constant $\lambda > 0$ such that*

$$\mathcal{W}_1(\mu, \pi) \leq \sqrt{2 \text{KL}(\mu \| \pi) / \lambda}$$

holds for every $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

Assumption 3. *There exists a constant $B > 0$, uniform in θ , such that the feature map satisfies $\|\Phi_\theta(x)\|_{\mathcal{H}_0} \leq B$ for all $x \in \mathbb{R}^d$ and $\theta \in \Theta$, and such that $\nabla \Phi_\theta$ is continuous and obeys $\|\nabla \Phi_\theta(x)\|_{\mathcal{H}}^2 = \sum_{i=1}^d \|\partial_i \Phi_\theta(x)\|_{\mathcal{H}_0}^2 \leq B^2$ for all $x \in \mathbb{R}^d$ and $\theta \in \Theta$.*

4.1 Convergence analysis of SVGD under Talagrand’s inequality

In the following, we denote

$$\mathcal{F}(\mu) := \text{KL}(\mu \| \pi).$$

and the analysis rests on the following fundamental descent inequality. Fix a kernel parameter $\theta \in \Theta$ for which Assumption 3 holds, and define the push-forward measure

$$\tilde{\mu} = (I + \gamma g) \# \mu$$

for an arbitrary field $g \in \mathcal{H}$. Under Assumptions 1 and 3, let $\gamma, B > 0$ and $\alpha > 1$, and let $g \in \mathcal{H}_\theta$ satisfy the step-size restriction $\gamma \|g\|_{\mathcal{H}_\theta} \leq \frac{\alpha-1}{\alpha B}$. Then Proposition 3.1 of [32] gives

$$\mathcal{F}(\tilde{\mu}) \leq \mathcal{F}(\mu) + \gamma \langle \psi_\theta^\mu, g \rangle_{\mathcal{H}_\theta} + \frac{\gamma^2 K}{2} \|g\|_{\mathcal{H}_\theta}^2 \quad (6)$$

where $K = (\alpha^2 + L)B$. Specializing to the iterates (3) generated under a fixed kernel parameter $\theta \in \Theta$ then yields the per-step descent condition [32, Theorem 3.2],

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2}\right) \text{KSD}_\theta^2(\mu_n | \pi), \quad (7)$$

provided that

$$\gamma \leq (\alpha - 1) \left(\alpha B (1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1}. \quad (8)$$

The crux in establishing (7) is verifying the step-size restriction $\gamma \|\psi_{\theta}^{\mu_n}\|_{\mathcal{H}_{\theta}} \leq \frac{\alpha-1}{\alpha B}$, which follows from Talagrand's inequality (Assumption 2) and licenses the application of (6) with the choice $g = \psi_{\theta}^{\mu_n}$. Summing the descent condition (7) then yields the vanishing of the KSD along the iterates,

$$\lim_{n \rightarrow \infty} \text{KSD}_{\theta}^2(\mu_n|\pi) = 0,$$

because the series is summable, $\sum_{n=0}^{\infty} \text{KSD}_{\theta}^2(\mu_n|\pi) \leq c_{\gamma}^{-1} \mathcal{F}(\mu_0)$, with $c_{\gamma} = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$.

When the kernel parameter is instead chosen adaptively as a sequence (θ_n) , the descent condition (7) generalizes to

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) \text{KSD}_{\theta_n}^2(\mu_n|\pi).$$

4.2 Convergence analysis of Ad-SVGd.

Suppose that the adaptive SVGd iteration admits the idealized form

$$\begin{aligned} \mu_{n+1} &= (I + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n, \\ \theta_n &\in \operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi), \end{aligned} \tag{9}$$

so that the selected direction attains the maximal discrepancy over the class,

$$\|\psi_{\theta_n}^{\mu_n}\|_{\mathcal{H}_{\theta_n}} = \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n|\pi).$$

This exact formulation is available precisely when the maximization of the KSD over the kernel parameter admits a (unique) closed-form solution. Such a closed form exists in the multiple-kernel SVGd framework [1], which parameterizes the kernel as a convex combination of base kernels,

$$k_{\theta}(x, y) = \sum_{i=1}^N \theta_i k_i(x, y) \quad \text{with} \quad \sum_{i=1}^N \theta_i = 1.$$

In the general setting, by contrast, the exact maximizer θ_n in (9) is unavailable and must be approximated. Throughout, we assume that the argmax set is nonempty, $\operatorname{argmax}_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) \neq \emptyset$ for all $n \in \mathbb{N}$.

The following descent condition follows directly from Theorem 3.2 of [32]:

Lemma 1. *Suppose that Assumptions 1-3 are satisfied. For any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}}) \right)^{-1},$$

there exists $c_{\gamma} > 0$ such that for all $n \in \mathbb{N}$

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi),$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (9).

Corollary 2. *Under the same assumptions as in Lemma 1 it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n|\pi) = 0$.*

The analysis is more delicate when the KSD maximization over the kernel parameter can only be solved approximately. Assume that $\max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu|\pi) < \infty$ for every $\mu \in \mathcal{P}(\mathbb{R}^d)$, and consider the alternating scheme

$$\begin{aligned}\theta_n &= \Psi_n(\theta_{n-1}, \mu_n), \\ \mu_{n+1} &= (\text{Id} + \gamma \psi_{\theta_n}^{\mu_n})_{\#} \mu_n\end{aligned}\tag{10}$$

for a sequence of iterative update rules $\Psi_n : \Theta \times \mathcal{P}(\mathbb{R}^d) \rightarrow \Theta$ designed to increase $\text{KSD}_{\theta}(\mu_n | \pi)$. In Algorithm 1, Ψ_n is instantiated as the gradient-ascent update on the KSD; as noted above, any alternative iterative optimizer may be substituted. Our sole requirement on the update rule is the following asymptotic accuracy condition.

Assumption 4. *There exists a nonnegative sequence $(\varepsilon_n)_{n \in \mathbb{N}}$ with $\lim_{n \rightarrow \infty} \varepsilon_n = 0$ such that the suboptimality of the parameter update is controlled,*

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) - \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \varepsilon_n \quad \text{for all } n \in \mathbb{N}.$$

Under this assumption, we obtain the following convergence guarantee.

Theorem 3. *Suppose that Assumptions 1-3 are satisfied. Under Assumption 4, for any $\alpha > 1$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

it holds that $\lim_{n \rightarrow \infty} \max_{\theta \in \Theta} \text{KSD}_{\theta}(\mu_n | \pi) = 0$, where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10).

Proof. Theorem 3.2 of [32] yields the descent inequality

$$\mathcal{F}(\mu_{n+1}) \leq \mathcal{F}(\mu_n) - c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi)$$

The sequence $(\mathcal{F}(\mu_n))_{n \in \mathbb{N}}$ is nonincreasing and bounded below by zero; hence it converges and is, in particular, Cauchy. It follows that

$$c_{\gamma} \text{KSD}_{\theta_n}^2(\mu_n | \pi) \leq \mathcal{F}(\mu_n) - \mathcal{F}(\mu_{n+1}) \rightarrow 0$$

as $n \rightarrow \infty$. Invoking Assumption 4, the maximal discrepancy is bounded by the achieved discrepancy plus the suboptimality gap, whence

$$\max_{\theta \in \Theta} \text{KSD}_{\theta}^2(\mu_n | \pi) \leq \text{KSD}_{\theta_n}^2(\mu_n | \pi) + \varepsilon_n \rightarrow 0$$

as $n \rightarrow \infty$. □

Under an additional Stein logarithmic Sobolev inequality, the convergence of Ad-SVGD can be strengthened from vanishing KSD to convergence in the KL divergence. For specific decay profiles of ε_n we obtain explicit iteration-complexity bounds for the mean-field limiting system (10). This setting further permits an explicit characterization of the asymptotic bias of Ad-SVGD when the KSD maximization is solved only to a fixed $\bar{\varepsilon}$ -accuracy at each iteration; the details are deferred to Section B. We stress that the inequality is required to hold for only a single kernel in the class $\{k_{\theta}\}$, which affords considerable flexibility in verifying the condition.

5 Numerical Experiments

In this section we evaluate Ad-SVGD on two benchmark problems: a one-dimensional Gaussian mixture model and a linear inverse problem governed by an ordinary differential equation. Complementary experiments isolating the effect of increasing dimension are reported in Section A, and Section 5.4 quantifies the advantage of Ad-SVGD over median-heuristic SVGD on a Bayesian logistic regression model previously employed in the SVGD literature.

5.1 Kernel parameterization

SVGD is most frequently instantiated with kernels of the form

$$k_h(x, y) = \exp\left(-\frac{\|x - y\|_p^p}{h}\right),$$

where $\|\cdot\|_p$ is the p -norm on $\mathbb{R}^d$ [e.g. 18, 2, 8]. We concentrate on strategies for selecting the scalar parameter h , termed the kernel *bandwidth*. The standard heuristic prescribes $h = \frac{\text{med}^p}{\log(M-1)}$, where med is the median pairwise distance among the current particles; this closed-form choice is designed to equilibrate the magnitudes of the two terms in the SVGD update (3) [18].

To exploit the expressivity of the adaptive method, we adopt product kernels of the form

$$k_h(x, y) = \prod_{i=1}^d \exp\left(-\frac{|x_i - y_i|^p}{h_i}\right)$$

parameterized by a bandwidth vector $h = (h_1, \dots, h_d)$, thereby assigning a distinct, dimension-specific bandwidth to each coordinate. The derivatives required to instantiate Algorithm 1 for these kernels (cf. Equations (4) and (5)) are available in closed form. For comparison, we also evaluated a coordinatewise variant of the median heuristic that assigns each dimension the naive per-coordinate median; this variant performed poorly and exhibited variance collapse.

Beyond the present experiments, the generality of Ad-SVGD accommodates more scalable kernel families in higher-dimensional regimes. In particular, spectral or low-rank kernel parameterizations can adapt to the dominant directions of variability without the cost of a full product kernel that scales unfavorably with dimension, whereas Matérn or mixed-product constructions afford additional control over smoothness and anisotropy.

5.2 Toy example

We first consider the one-dimensional benchmark of Liu and Wang [18], namely the two-component Gaussian mixture $\pi = \frac{1}{3}\mathcal{N}(-2, 1) + \frac{2}{3}\mathcal{N}(2, 1)$. The particle ensemble, of size $M \in \{50, 200, 500\}$, is initialized by i.i.d. draws from $\mathcal{N}(0, 1)$. We iterate SVGD for 10^4 steps at a step size of 1, using the kernel family introduced above with $p = 1$ and a grid of fixed bandwidths h . Sample quality is assessed by the Wasserstein 1-distance $\mathcal{W}_1$, evaluated via the

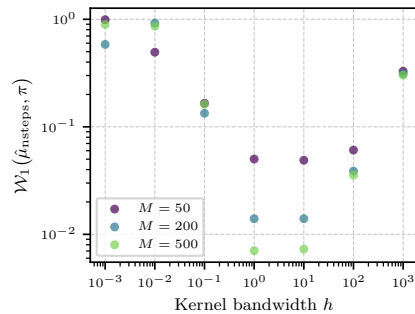


Figure 1: Terminal Wasserstein 1-distances $\mathcal{W}_1$ for the one-dimensional example, obtained by SVGD across a range of fixed bandwidths h .

closed-form univariate expression $\mathcal{W}_1(\mu, \nu) = \int_{\mathbb{R}} |F_{\mu}(x) - F_{\nu}(x)| dx$ in terms of the cumulative distribution functions [29], against an exact reference sample of size 10^5 . The outcomes are reported in Figure 1: the algorithm attains low error only for bandwidths within a narrow interval. This pronounced sensitivity of the terminal error to h demonstrates that a principled bandwidth-selection strategy is indispensable.

5.3 Linear inverse problem based on ODE

The following example is adapted from [38, Example 2.1]. We consider the one-dimensional two-point boundary value problem

$$\begin{cases} -f''(s) + f(s) = u(s) & \text{for } s \in (0, 1) \\ f(s) = 0 & \text{for } s \in \{0, 1\} \end{cases} \quad (11)$$

together with the associated inverse problem of recovering the source term $u(\cdot) \in L^2([0, 1])$ from finitely many noisy point observations of the solution $f \in H^2([0, 1]) \cap H_0^1([0, 1])$. The observation model is

$$y = \Phi(u) + \varepsilon \in \mathbb{R}^{N_{\text{obs}}},$$

where $\varepsilon \in \mathbb{R}^{N_{\text{obs}}}$ denotes additive observational noise and the forward operator $\Phi : L^2([0, 1]) \rightarrow \mathbb{R}^{N_{\text{obs}}}$ is the composition $\mathcal{O} \circ H^{-1}$, with solution operator H^{-1} inverting $H(f) = -f'' + f$ and observation operator $\mathcal{O}(f) = (f(s_1), \dots, f(s_{N_{\text{obs}}}))^{\top} \in \mathbb{R}^{N_{\text{obs}}}$ evaluating f at the N_{obs} equispaced nodes $s_k = \frac{k}{N_{\text{obs}}}$, $k = 1, \dots, N_{\text{obs}}$.

For the Bayesian formulation, we place a Gaussian process (GP) prior on u specified through the truncated Karhunen-Loève (KL) expansion

$$u(\cdot, x) = Ax = \sum_{k=1}^{N_x} x_k \psi_k,$$

with basis functions $\psi_k(s) = \sqrt{2} \sin(\pi k s)$ and independent coefficients $x_k \sim \mathcal{N}(0, \lambda_k)$, where the eigenvalues decay as $\lambda_k = 50k^{-2}$. Inference thus reduces to estimating the vector of KL coefficients $x = (x_1, \dots, x_{N_x})^{\top} \in \mathbb{R}^{N_x}$ under the Gaussian prior $\mathcal{N}(0, \Gamma_0)$ with $\Gamma_0 = \text{diag}(\lambda_1, \dots, \lambda_{N_x})$. Under the Gaussian noise model $\varepsilon \sim \mathcal{N}(0, \Gamma)$, Bayes' rule yields the posterior density

$$\pi(x) \propto \exp \left(-\frac{1}{2} \|\Gamma^{-1/2}(y - \Phi Ax)\|^2 - \frac{1}{2} \|\Gamma_0^{-1/2} x\|^2 \right), \quad x \in \mathbb{R}^{N_x}.$$

In the implementation, the operator H is replaced by a finite-difference discretization of Equation (11) on a uniform grid of mesh size 2^{-8} , and we take the system to be fully observed, i.e. $N_{\text{obs}} = 2^8$. The KL expansion of u is truncated at $N_x = 16$ terms, and the noise covariance is set

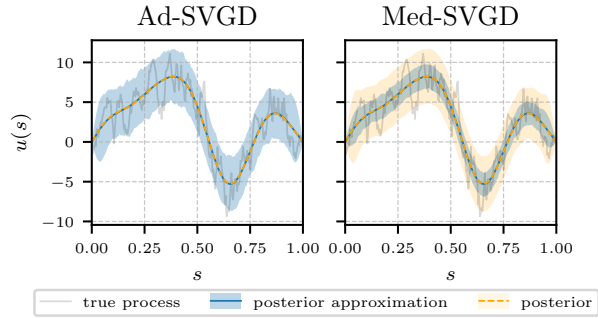


Figure 2: Gaussian-process reconstruction for the ODE-based inverse problem, showing the posterior mean and the associated 90% credible interval.

Under the Gaussian noise model $\varepsilon \sim \mathcal{N}(0, \Gamma)$, Bayes' rule yields the posterior density

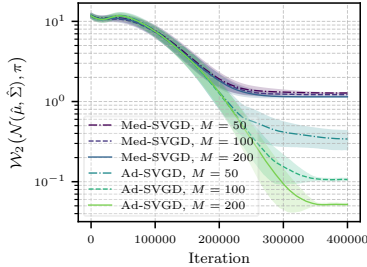
to $\Gamma = 10^{-3} \text{Id}_{N_{\text{obs}}}$. Reference observations $\bar{y} \in \mathbb{R}^{N_{\text{obs}}}$ are generated by sampling $\bar{x} \sim \mathcal{N}(0, \Gamma_0)$ and setting $\bar{y} = \Phi A \bar{x}$.

We benchmark median-heuristic SVGD (denoted Med-SVGd) against Ad-SVGd for ensemble sizes $M \in \{50, 100, 200\}$, employing the kernel family of Section 5.1 with $p = 1$. Both algorithms are run for $4 \cdot 10^5$ iterations at a particle step size of 10^{-3} , combined—following Liu and Wang [18]—with an AdaGrad-type rule for adaptive step-size control. For Ad-SVGd, the bandwidth is updated by gradient ascent at step size 10^{-5} , performed only once per 100 particle updates; under this schedule the wall-clock cost of Ad-SVGd is indistinguishable from that of Med-SVGd.

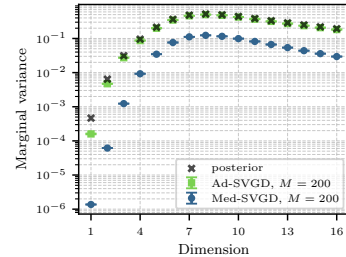
Figure 2 displays the GP reconstruction for a representative seed. Both methods recover the posterior mean accurately, but only Ad-SVGd faithfully reproduces the posterior uncertainty. To quantify approximation quality, we adopt the Wasserstein 2-distance $\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)$ between the posterior π and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ induced by the empirical mean $\hat{\mu}$ and empirical covariance $\hat{\Sigma}$ of the particle ensemble [29, Section 3.2]. Because π is itself multivariate normal, this distance admits the closed form

$$\mathcal{W}_2(\mathcal{N}(\hat{\mu}, \hat{\Sigma}), \pi)^2 = \|\hat{\mu} - \mu_\pi\|^2 + \text{trace} \left(\hat{\Sigma} + \Sigma_\pi - 2(\hat{\Sigma}^{1/2} \Sigma_\pi \hat{\Sigma}^{1/2})^{1/2} \right),$$

where μ_π and Σ_π denote the posterior mean and covariance. As a complementary diagnostic, we compare the marginal variances of the terminal particle distribution against those of the posterior. Figure 3 aggregates the results over 56 independent random seeds (the posterior covariance is invariant to $\bar{y}$). Consistent with the preceding observations, Ad-SVGd attains a closer posterior approximation than Med-SVGd, which systematically underestimates the uncertainty. Moreover, and in contrast to Med-SVGd, the approximation error of Ad-SVGd decreases monotonically as the ensemble size is increased beyond 50. Figure 4 characterizes the bandwidth trajectories produced by Ad-SVGd: the coordinatewise bandwidths equilibrate appreciably faster than the approximation error, exhibit pronounced dispersion across coordinates, and are negatively correlated with the corresponding marginal variances.



(a) Convergence of the approximation error.

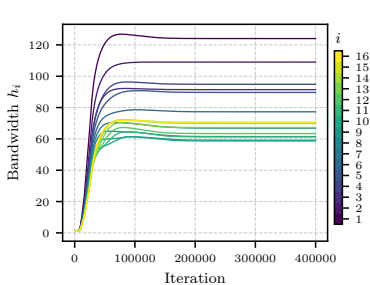


(b) Terminal marginal variances.

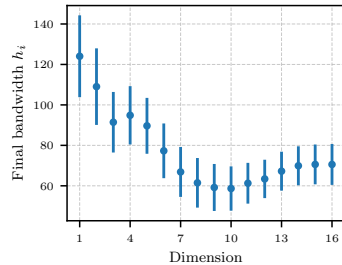
Figure 3: Aggregated results for the ODE-based inverse problem under Med-SVGd and Ad-SVGd (mean and 95% confidence interval over 56 random seeds).

5.4 Bayesian logistic regression

Following Liu and Wang [18] and Liu et al. [21], we consider a Bayesian logistic regression (BLR) model on the Covtype data set [4]. The regression weights w are assigned a hierarchical Gaussian



(a) Per-coordinate bandwidth trajectories (mean).



(b) Terminal bandwidths (mean and 95% confidence interval).

Figure 4: Behavior of the Ad-SVGD bandwidth parameter for the ODE-based inverse problem, aggregated over 56 random seeds.

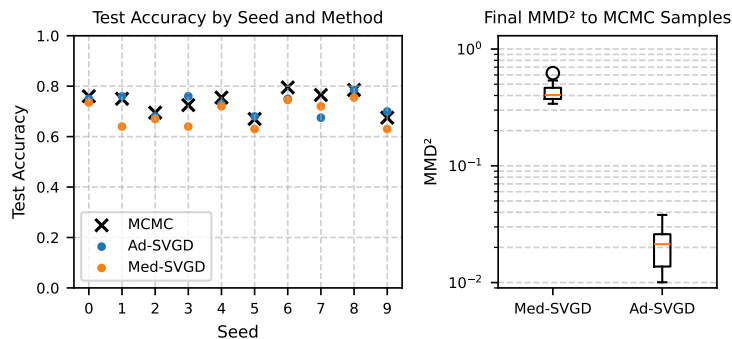


Figure 5: Approximation quality of the SVGD particle ensembles for Bayesian logistic regression, quantified by test-set prediction accuracy (left) and by the squared maximum mean discrepancy MMD^2 to the reference samples (right).

prior $w \mid \alpha \sim \mathcal{N}(0, \alpha^{-1})$ with precision hyperparameter $\alpha \sim \text{Gamma}(1, 0.01)$, and the inferential target is the joint posterior of the augmented parameter $x = [w, \alpha]$. As in Liu et al. [21], we obtain a high-fidelity reference for the posterior via the No-U-Turn Sampler (NUTS) of Hoffman and Gelman [13], against which the approximation quality of Med-SVGD and Ad-SVGD is assessed. Again following Liu et al. [21], we evaluate the methods on random subsets of the data set of size 1000, aggregating results over 10 independent draws. Classification performance is measured by the test-set prediction accuracy induced by the particle mean. As Figure 5 shows, Ad-SVGD dominates Med-SVGD on nearly every seed, simultaneously attaining higher test accuracy and closer agreement with the MCMC reference.

As a direct measure of posterior fidelity, the same figure reports the squared maximum mean discrepancy (MMD^2) [25, Section 3.5] to the MCMC samples. The two methods differ markedly in their ability to capture the posterior. This is corroborated by Figure 7, which contrasts the

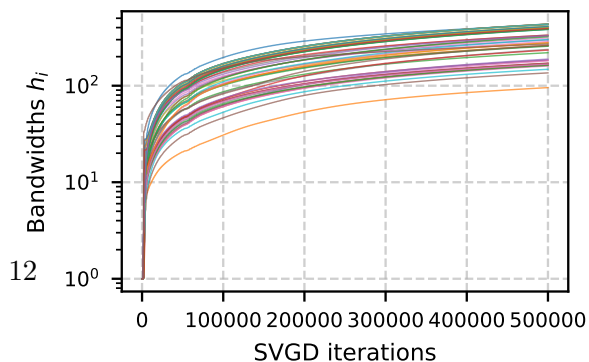


Figure 6: Per-coordinate bandwidth trajectories for the PDE example.

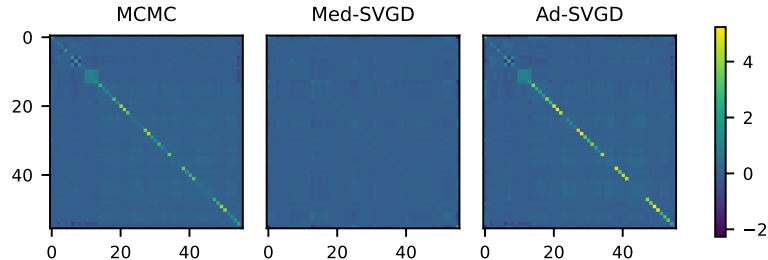


Figure 7: Terminal SVGD particle covariance matrices compared to the MCMC reference (mean over 10 seeds).

terminal particle covariance matrices with the MCMC reference, averaged over the 10 seeds (the visualization follows Liu et al. [21]): Med-SVGd severely underestimates the posterior uncertainty, whereas Ad-SVGd yields a substantially closer approximation. Finally, Figure 6 depicts the per-coordinate bandwidth trajectories.

Starting from an uninformative initialization (unit bandwidth in every coordinate), Ad-SVGd recovers coordinate-appropriate bandwidths that differ from the initialization by several orders of magnitude.

6 Limitations and outlook

The principal limitation of the analysis is its dependence on Assumption 4. Ongoing work seeks to identify conditions under which the alternating gradient-ascent scheme of Algorithm 1, as instantiated in our experiments, provably satisfies this assumption. While we do not currently establish Assumption 4 for that scheme, the resulting implementation attains strong empirical performance.

The present analysis is confined to the original SVGD dynamics. A natural direction is to compose the proposed adaptive kernel-selection mechanism with recent algorithmic variants, including sliced SVGD [10], Grassmann SVGD [21], and Stein transport [27].

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A More numerical experiments

A.1 Gaussian mixture models

We revisit the setup of Section 5.2. For that configuration, Figure 9 juxtaposes histograms of the terminal particle distribution with the target density for the fixed bandwidths $h \in \{0.001, 1, 1000\}$. We now compare Med-SVGD with Ad-SVGD, contrasting the terminal particle distributions produced by the two methods after 10^4 iterations at step size 1 across a range of ensemble sizes M .

Figure 8 reports the Wasserstein 1-distance between the terminal empirical measure and the target for

Figure 8: Approximation error for the one-dimensional Gaussian mixture.

Med-SVGD and Ad-SVGD at $M = 10, 20, 50, 100, 200, 500$. As anticipated, the approximation error decreases with N for both methods, and the two are essentially indistinguishable here, each attaining a Wasserstein distance below 0.01 at $N = 500$.

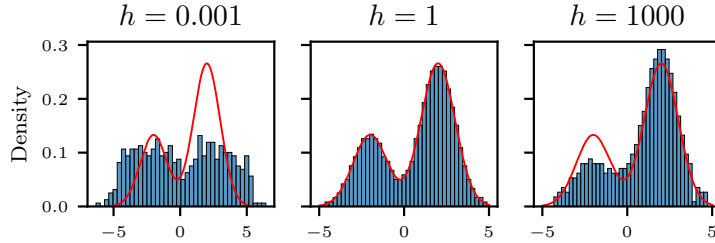
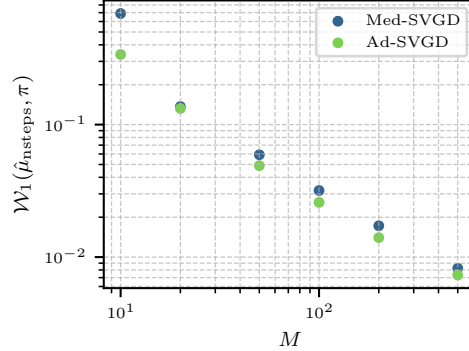


Figure 9: Histograms of the terminal particle set for the fixed bandwidths $h = 0.001, 1, 1000$; the target density π is overlaid in red for reference.

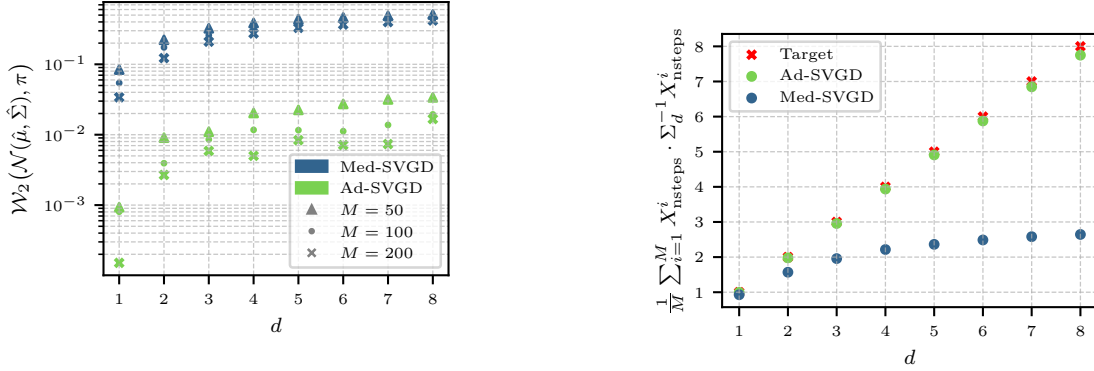
In the one-dimensional case, both methods approximate the target accurately. The adaptive bandwidth-selection strategy performs well but confers no material advantage over the median heuristic in this low-dimensional regime.

A.2 Scaling dimension: multivariate normal distribution

To probe the dimensional scaling, we consider the Gaussian targets $\pi_d = \mathcal{N}(0, \Sigma_d)$ with anisotropic covariance $\Sigma_d = \text{diag}(1, \frac{1}{2}, \dots, \frac{1}{d})$ for $d \in \{2, \dots, 8\}$. The particle ensemble is initialized from the product measure $\mathcal{N}(0, \frac{1}{d})^{\otimes d}$, and each algorithm is run for 10^4 iterations at step size 0.1; where numerical stability so required, the step size was reduced and the iteration count increased commensurately.

We compare the approximation quality of the terminal particle ensembles from Med-SVGD and Ad-SVGD as d increases. Since the Wasserstein 2-distance between two Gaussians admits a

closed form [29], we adopt as sample-quality metric the Wasserstein 2-distance between the target π_d and the Gaussian $\mathcal{N}(\hat{\mu}, \hat{\Sigma})$ fitted to the empirical mean $\hat{\mu}$ and covariance $\hat{\Sigma}$ of the particles [29, Section 3.2]. Consistent with the terminology of the Python Optimal Transport library [9] used for our computations, we refer to this quantity as the *Bures–Wasserstein distance*. Figure 10a traces this metric as a function of d for Med-SVGD and Ad-SVGD at $M = 50, 100, 200$. Ad-SVGD outperforms Med-SVGD by a substantial margin at every value of d .



(a) Bures–Wasserstein approximation error for $M \in \{50, 100, 200\}$.

(b) Mean χ^2 -test statistic for $M = 200$.

Figure 10: Results for multivariate normal targets of increasing dimension.

The Bures–Wasserstein distance is relatively insensitive to the small variances of the trailing components of π_d ; to expose the consequences of failing to capture the uncertainty in those components, we examine the quadratic-form statistic $X \cdot \Sigma_d^{-1} X$, which for $X \sim \mathcal{N}(0, \Sigma_d)$ is χ^2 -distributed with d degrees of freedom and hence has expectation d (cf. [34, p. 27]). We evaluate the empirical mean of this statistic over the $M = 200$ particles and compare it against its expectation d in Figure 10b. Beyond $d = 1$, the Med-SVGD statistic diverges increasingly from the true expectation as d grows, whereas the Ad-SVGD statistic remains close to d throughout. This demonstrates the failure of Med-SVGD to approximate higher-dimensional targets and the robustness of Ad-SVGD to increasing dimension.

Tables 1 and 2 report the terminal particle distributions in finer detail for $M = 200$ and varying d , tabulating the per-coordinate marginal variances of the particle ensemble alongside the target marginal variances listed in the first row; Table 1 corresponds to Med-SVGD and Table 2 to Ad-SVGD. Ad-SVGD reproduces the target marginal variances accurately across all coordinates, whereas Med-SVGD markedly underestimates the target uncertainty—corroborating the trends of Figure 10b.

Finally, we examine the marginal particle distributions for $M = 200$ and $d = 8$. To standardize the visualization, each particle coordinate is rescaled by the reciprocal of the corresponding target marginal standard deviation (equivalently, the i -th coordinate is multiplied by i), which maps every marginal of π_d onto the standard normal law. Figures 11 and 12 display the resulting normalized marginal histograms for $d = 8$ under Med-SVGD and Ad-SVGD, respectively, with the standard Gaussian density overlaid for reference. As before, Ad-SVGD captures every marginal accurately while Med-SVGD underestimates the target uncertainty. The same conclusion follows from Figures 13 and 14, which compare the quantiles of the normalized marginal particle distributions against those of the standard normal target.

Table 1: Terminal per-coordinate marginal variances for the d -dimensional targets under Med-SVGD with $M = 200$; the target marginal variances of π_d are listed for reference.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9285							
$d = 2$	0.7921	0.1943						
$d = 3$	0.6803	0.1625	0.0697					
$d = 4$	0.6089	0.1440	0.0593	0.0311				
$d = 5$	0.5532	0.1275	0.0526	0.0271	0.0157			
$d = 6$	0.5190	0.1190	0.0481	0.0243	0.0140	0.0089		
$d = 7$	0.4900	0.1122	0.0449	0.0228	0.0131	0.0081	0.0052	
$d = 8$	0.4753	0.1077	0.0430	0.0215	0.0122	0.0074	0.0047	0.0032

Table 2: Terminal per-coordinate marginal variances for the d -dimensional targets under Ad-SVGD with $M = 200$; the target marginal variances of π_d are listed for reference.

component	1	2	3	4	5	6	7	8
Target	1.0000	0.2500	0.1111	0.0625	0.0400	0.0278	0.0204	0.0156
$d = 1$	0.9953							
$d = 2$	0.9907	0.2472						
$d = 3$	0.9867	0.2459	0.1095					
$d = 4$	0.9881	0.2467	0.1095	0.0610				
$d = 5$	0.9840	0.2433	0.1096	0.0616	0.0392			
$d = 6$	0.9858	0.2459	0.1090	0.0611	0.0392	0.0269		
$d = 7$	0.9856	0.2463	0.1086	0.0613	0.0390	0.0269	0.0199	
$d = 8$	0.9691	0.2409	0.1085	0.0611	0.0390	0.0268	0.0196	0.0150

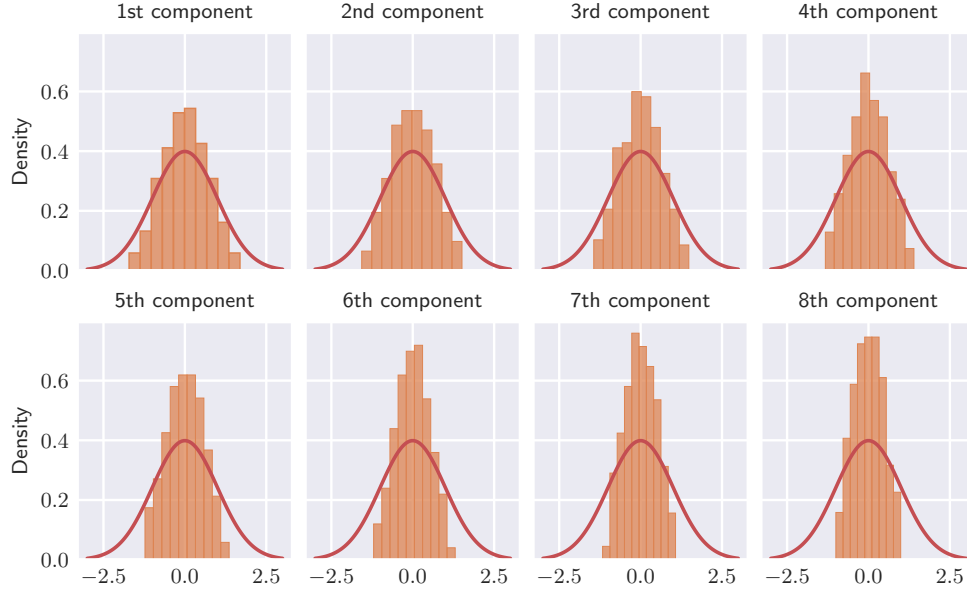


Figure 11: Coordinatewise histograms of the terminal particle set for the eight-dimensional target under Med-SVGD with $M = 200$, rescaled by the target marginal standard deviations; the standard Gaussian density is overlaid for reference.

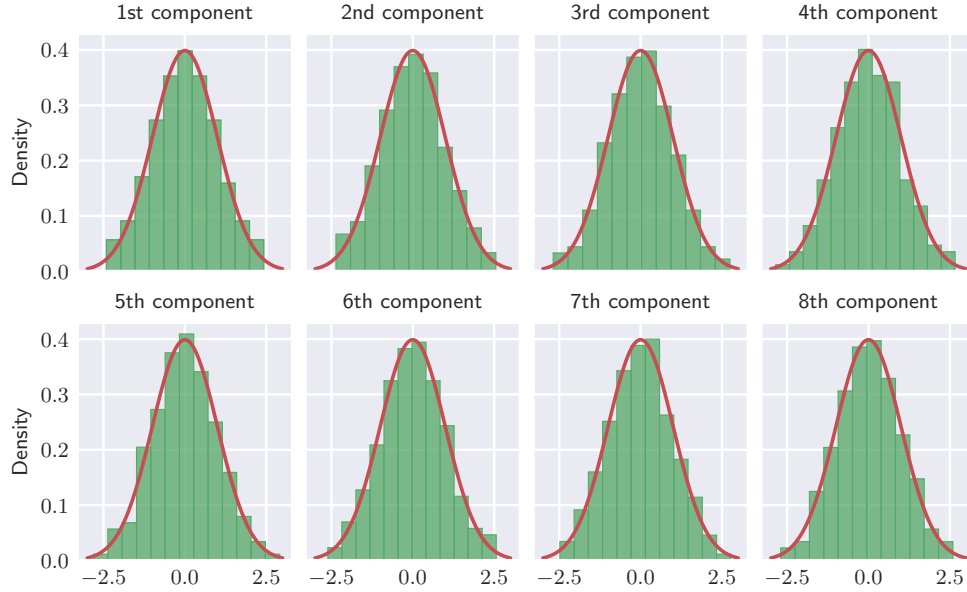


Figure 12: Coordinatewise histograms of the terminal particle set for the eight-dimensional target under Ad-SVGD with $M = 200$, rescaled by the target marginal standard deviations; the standard Gaussian density is overlaid for reference.

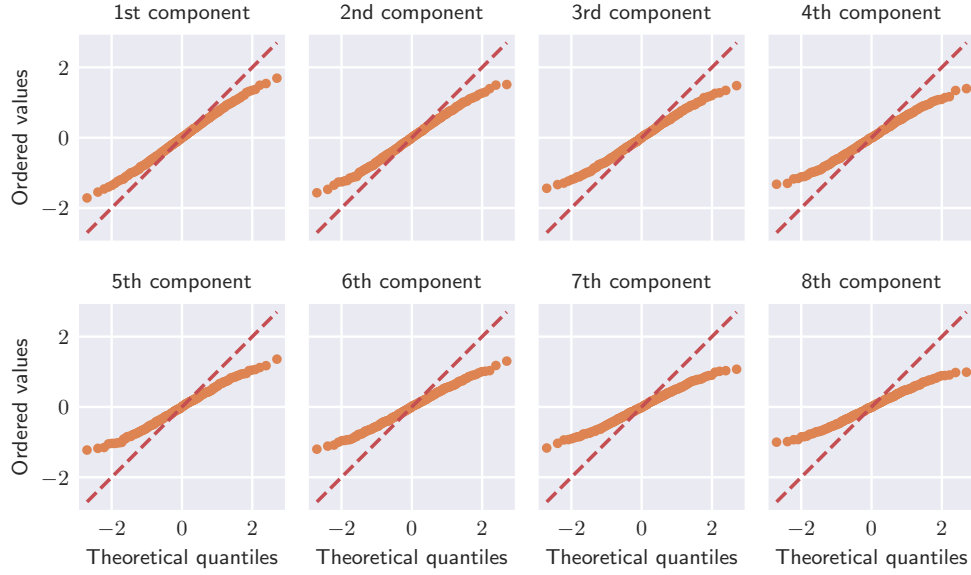


Figure 13: Q–Q plots of the rescaled terminal marginal particle distributions against the standard normal for the eight-dimensional target under Med-SVGd with $M = 200$; the identity line (slope 1 through the origin) is shown for reference.

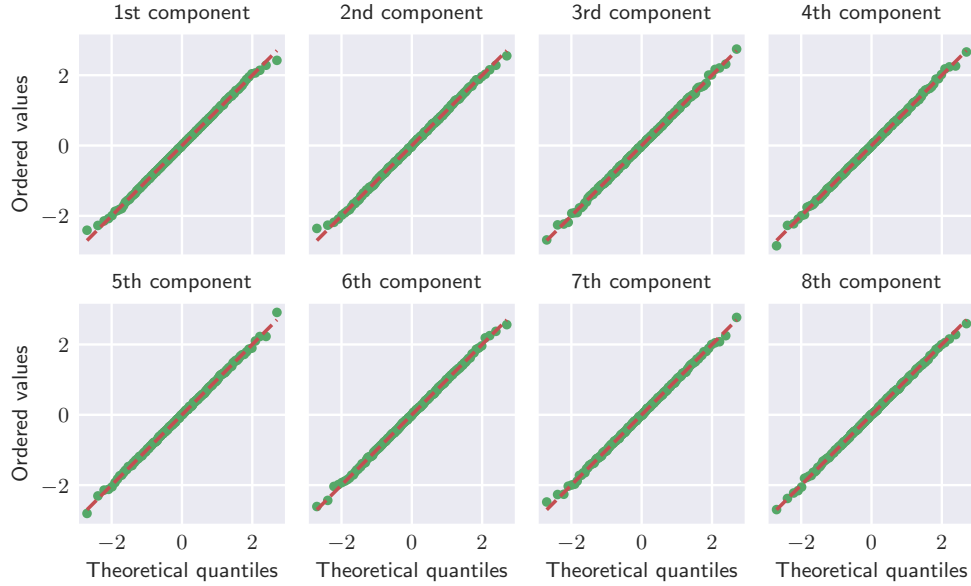


Figure 14: Q–Q plots of the rescaled terminal marginal particle distributions against the standard normal for the eight-dimensional target under Ad-SVGd with $M = 200$; the identity line (slope 1 through the origin) is shown for reference.

A.3 Gaussian process inference

We further assess the dimensional scaling of Ad-SVGD on the Gaussian process (GP) inference task of [31]. Consider a GP on $[0, 1]$ represented by the truncated Karhunen-Loève (KL) expansion $u(s, x) = \sum_{k=1}^{N_x} x_k \psi_k(s)$ with basis functions $\psi_k(s) = \sqrt{2} \sin(k\pi s)$, where $x = (x_1, \dots, x_{N_x})^\top$ collects independent Gaussian coefficients $x_k \sim \mathcal{N}(0, k^{-2})$. The process is observed at N_y equispaced points in $[0, 1]$, and the coefficients x_k are to be inferred. For fixed N_x and N_y this is a linear-Gaussian inverse problem with forward model $Y = AX + \varepsilon$, where the k -th column of $A \in \mathbb{R}^{N_y \times N_x}$ is $(\psi_k(s_1), \dots, \psi_k(s_{N_y}))^\top$ and $s_i = \frac{i}{N_y}$ for $i = 1, \dots, N_y$. The prior is $X \sim \mathcal{N}(0, \Sigma)$ with diagonal covariance Σ of entries k^{-2} , $k = 1, \dots, N_x$, and the noise is independent Gaussian, $\varepsilon \sim \mathcal{N}(0, I_{N_y})$. Reference observations $\bar{y} \in \mathbb{R}^{N_y}$ are generated by drawing $\bar{x} \sim \mathcal{N}(0, \Sigma)$ and setting $\bar{y} = A\bar{x}$.

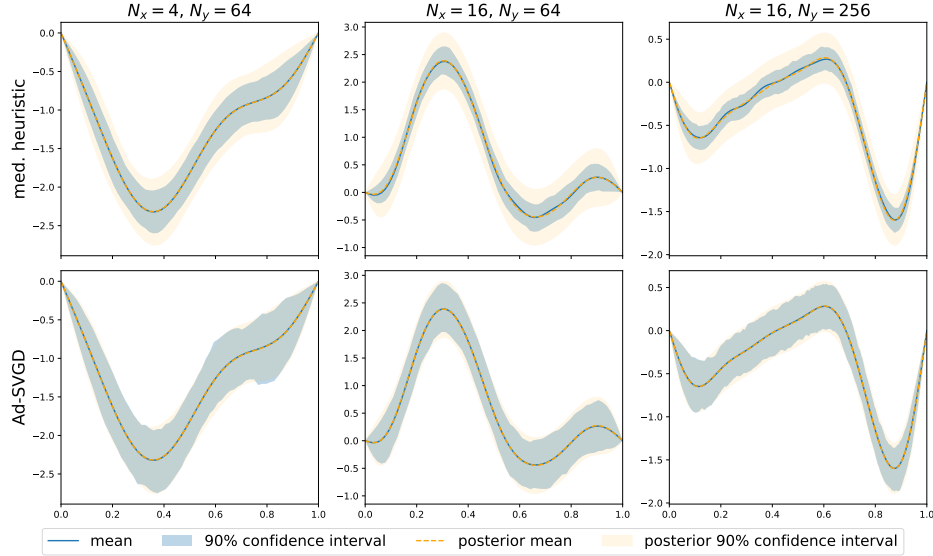


Figure 15: Estimated processes under the median heuristic and Ad-SVGD with $M = 100$ particles across several model configurations, compared against the posterior.

We apply SVGD to sample from the resulting posterior $X | Y = \bar{y}$ and contrast the median heuristic with the adaptive approach. Median-heuristic SVGD uses the isotropic kernel $k(x, y) = \exp(-\|x - y\|_1/h)$, whereas Ad-SVGD uses the product kernel $k(x, y) = \prod_{i=1}^{N_x} \exp(-|x_i - y_i|/h_i)$ with a distinct bandwidth h_i per coordinate, $h = (h_1, \dots, h_{N_x})$. Following [18], step-size control in the $N_x = 16$ configurations is handled by an AdaGrad-type rule.

Figure 15 overlays the estimated processes on the posterior for the three configurations $(N_x, N_y) \in \{(4, 64), (16, 64), (16, 256)\}$. As the dimension grows, median-heuristic SVGD underestimates the posterior variance, whereas Ad-SVGD retains a close approximation; this behavior is stable across the number of observations. Table 3 quantifies the comparison via the trace of the particle covariance relative to the true posterior: the median heuristic severely underestimates the uncertainty, while Ad-SVGD recovers the variance far more accurately.

N_x	4	8	16	16	16
N_y	64	64	64	128	256
theoretical	0.056	0.083	0.086	0.051	0.029
med. heuristic	0.026	0.023	0.022	0.012	0.006
Ad-SVGD	0.055	0.072	0.074	0.044	0.026

Table 3: Trace of the terminal particle covariance ($M = 100$ particles) compared against the theoretical posterior across configurations, averaged over 25 runs.

B Refined theoretical analysis

The convergence result can be sharpened under a Stein log-Sobolev inequality, which bounds the KL divergence by the KSD. This inequality was introduced by Duncan, Nüsken, and Szpruch [8], who also examined conditions for its validity. A notable feature of the adaptive setting is that the condition can be weakened: it suffices that a single kernel in the class satisfy the log-Sobolev inequality. Precisely, we impose the following assumption.

Assumption 5. *For a given kernel class $\{k_\theta \mid \theta \in \Theta\}$, the target $\pi \in \mathcal{P}_1(\mathbb{R}^d)$ satisfies the generalized Stein log-Sobolev inequality with constant $\lambda > 0$:*

$$2\lambda \text{KL}(\mu \parallel \pi) \leq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu \parallel \pi)$$

for all $\mu \in \mathcal{P}_1(\mathbb{R}^d)$.

This property upgrades the guarantee to convergence in the KL divergence. Crucially, it furnishes an explicit error bound on $\mathcal{F}(\mu_n)$ even under the weaker requirement $\varepsilon_n \leq \bar{\varepsilon}$, i.e. without demanding $\varepsilon_n \rightarrow 0$.

Theorem 4. *Suppose that Assumptions 1 to 3 and 5 are satisfied. For any $\alpha > 1$ and $\gamma > 0$ with*

$$\gamma \leq (\alpha - 1) \left(\alpha B(1 + \|\nabla V(0)\| + L \int_{\mathbb{R}^d} \|x\| d\pi(x) + L \sqrt{\frac{2\mathcal{F}(\mu_0)}{\lambda}} \right)^{-1},$$

and

$$\rho := 1 - 2\lambda c_\gamma \in (0, 1) \quad \text{where} \quad c_\gamma := \gamma \left(1 - \frac{\gamma B(\alpha^2 + L)}{2} \right),$$

it holds that

$$\mathcal{F}(\mu_{n+1}) \leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n,$$

where $(\mu_n)_{n \in \mathbb{N}}$ is generated by (10) and $\varepsilon_n \geq \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \parallel \pi) - \text{KSD}_{\theta_n}^2(\mu_n \parallel \pi)$. In particular, if

1. $\varepsilon_n \leq \bar{\varepsilon}$ for all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{c_\gamma \bar{\varepsilon}}{1 - \rho}$$

for all $n \in \mathbb{N}$.

2. $\varepsilon \leq c_\varepsilon q^n$ for some $q \in (0, 1)$, $c_\varepsilon > 0$ and all $n \in \mathbb{N}$. Under $\rho \neq q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\max\{\rho, q\}^n)$$

and under $\rho = q$ we have

$$\mathcal{F}(\mu_n) \in \mathcal{O}(\tilde{q}^n)$$

for all $\rho < \tilde{q} < 1$.

3. $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$ for $c_\varepsilon, p > 0$ and all $n \in \mathbb{N}$, then

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}$$

for all $n \in \mathbb{N}$.

Proof. Similar to the proof of Theorem 3, under Assumption 5 and the conditions on α, γ , we have

$$\begin{aligned} \mathcal{F}(\mu_{n+1}) &\leq \mathcal{F}(\mu_n) - c_\gamma \max_{\theta \in \Theta} \text{KSD}_\theta^2(\mu_n \mid \pi) + c_\gamma \varepsilon_n \\ &\leq \rho \mathcal{F}(\mu_n) + c_\gamma \varepsilon_n, \end{aligned}$$

where $c_\gamma = \gamma(1 - \frac{\gamma B(\alpha^2 + L)}{2}) > 0$. Unrolling this linear recursion over $n \in \mathbb{N}$ yields the general error bound

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^{n-1-k} \varepsilon_k.$$

We treat the three prescribed decay regimes for ε_n in turn:

1. For the uniformly bounded case $\varepsilon_n \leq \bar{\varepsilon}$, the first claim follows by summing the geometric series

$$\sum_{k=0}^{n-1} \rho^{n-1-k} \leq \sum_{k=0}^{\infty} \rho^k = \frac{1}{1-\rho}.$$

2. For the geometrically decaying case $\varepsilon_n \leq c_\varepsilon q^n$ with $q \in (0, 1)$ and $c_\varepsilon > 0$, we have

$$\mathcal{F}(\mu_n) \leq \rho^n \mathcal{F}(\mu_0) + c_\gamma c_\varepsilon \sum_{k=0}^{n-1} \rho^{n-1-k} q^k.$$

When $q > \rho$, summing the geometric series gives

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \sum_{k=0}^{n-1} q^{n-1-k} \rho^k = q^{n-1} \sum_{k=0}^{n-1} (\rho/q)^k \leq q^{n-1} \frac{1}{1-\rho/q} = q^n \frac{1}{q-\rho}$$

Symmetrically, when $q < \rho$, we bound directly

$$\sum_{k=0}^{n-1} \rho^{n-1-k} q^k = \rho^{n-1} \sum_{k=0}^{n-1} (q/\rho)^k \leq \rho^{n-1} \frac{1}{1-q/\rho} = \rho^n \frac{1}{(\rho-q)}.$$

In the boundary case $q = \rho$, we bound $q^k < \tilde{q}^k$ for any $q < \tilde{q} < 1$ and deduce the claim exactly as above.

3. For the polynomially decaying case $\varepsilon_n \leq \frac{c_\varepsilon}{(n+1)^p}$, we split the error bound on $\mathcal{F}(\mu_n)$ as

$$\begin{aligned}
\mathcal{F}(\mu_n) &\leq \rho^n \mathcal{F}(\mu_0) + c_\gamma \sum_{k=0}^{n-1} \rho^k \varepsilon_{n-1-k} \\
&= \rho^n \mathcal{F}(\mu_0) + c_\gamma \left(\sum_{k=0}^{\lfloor n/2 \rfloor} \rho^k \varepsilon_{n-1-k} + \sum_{k=\lfloor n/2 \rfloor + 1}^{n-1} \rho^k \varepsilon_{n-1-k} \right) \\
&\leq \rho^n \mathcal{F}(\mu_0) + \frac{2^p c_\gamma c_\varepsilon}{n^p} \frac{1}{1-\rho} + c_\gamma c_\varepsilon \frac{\rho^{\lfloor n/2 \rfloor}}{1-\rho}.
\end{aligned}$$

Here we have used the bounds $\varepsilon_{n-1-k} \leq \frac{c_\varepsilon}{(n-k)^p} \leq \frac{2^p c_\varepsilon}{n^p}$, valid for all $k \leq \lfloor n/2 \rfloor$, together with $\varepsilon_k \leq c_\varepsilon$ for all $k \in \mathbb{N}$.

□